ECA5802 IOE and AI Driven Systems

Exp no 4

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Simulation and Comparative Performance Evaluation of Zigbee, 6LoWPAN and NB-IoT Communication Protocols for IoE Applications

Objective:

To simulate and compare the performance of Zigbee, 6LoWPAN, and NB-IoT communication protocols by analysing Bit Error Rate (BER) with respect to communication distance.

Procedure :

1. Import NumPy, Matplotlib, and the erfc function from SciPy.
2. Define the communication distance range from 10 m to 1000m.
3. Set the transmission power, path loss, path loss exponent, and noise power for each protocol.
4. Calculate path loss, received power, SNR, and BER using the given equations.
5. Plot BER versus distance separately for Zigbee, 6LoWPAN, and NB-IoT.
6. Compare the plots to evaluate the effect of distance on communication performance

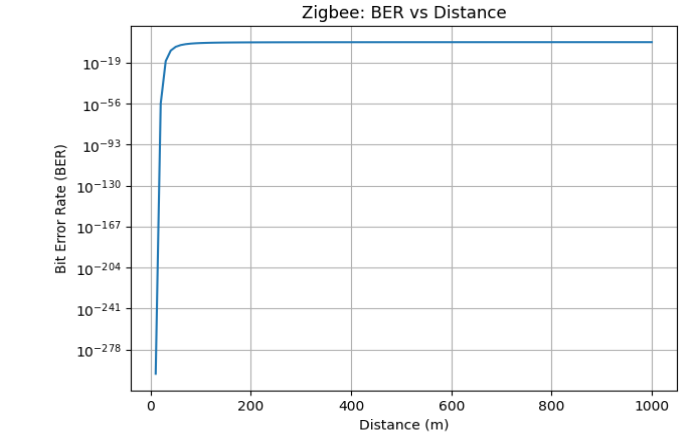
Tabulation :

N (dBm)	Zigbee (10m)	6LoWPAN (20m)	NB-IoT (400 m)
N = -100	0	1.29x10^-56	0.057
N = -95	7.30x10^-140	3.01x10^-19	0.188
N = -102	0	1.82x10^-88	0.0237

Output :

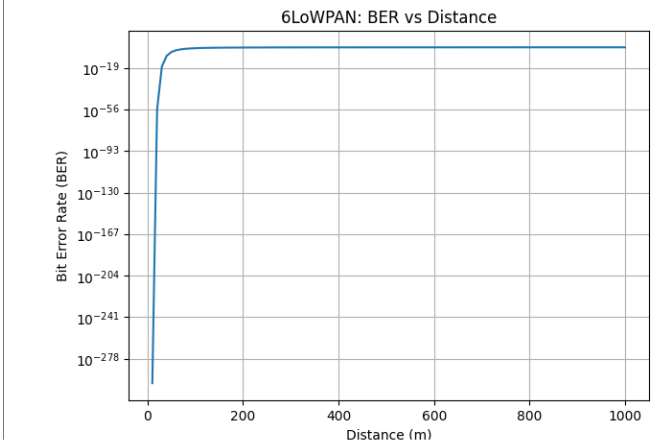
(i) N = -100

```
Zigbee BER at 10 m = 0.0
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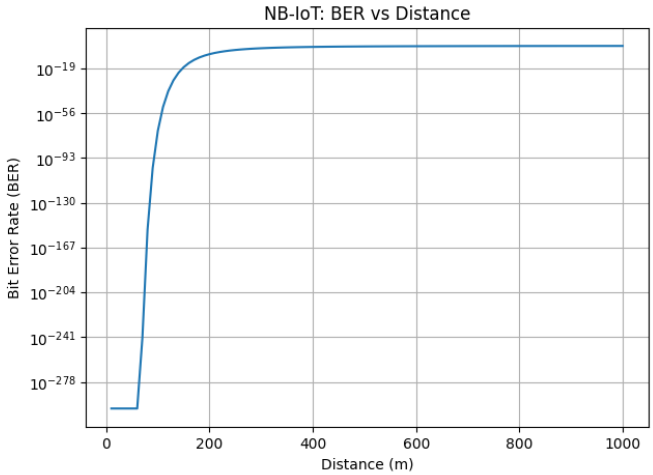


Distance (m)

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6LoWPAN BER at 20 m = 1.2984035196701857e-56
```

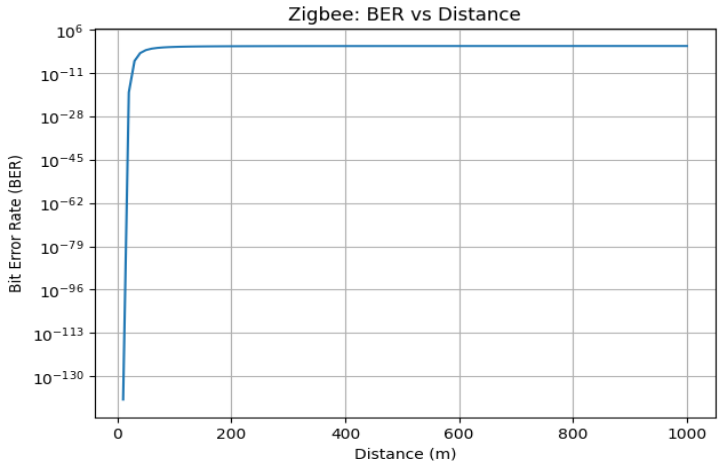


```
NB-IoT BER at 400 m = 0.05778341116369948
```

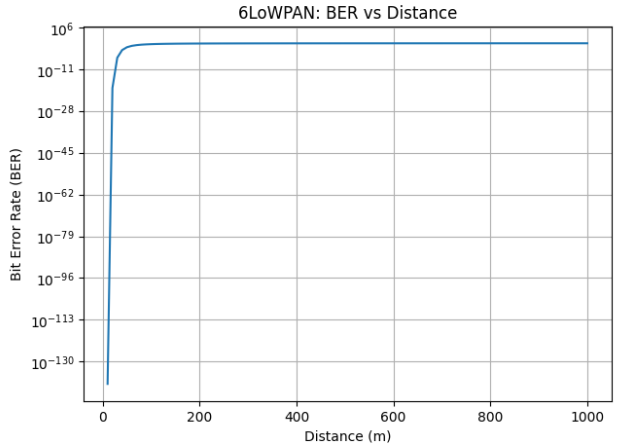


(ii) N = -95

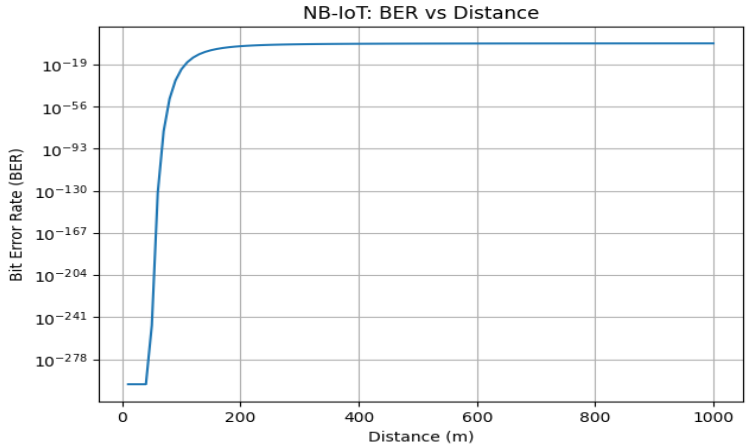
```
Zigbee BER at 10 m = 7.30696918464 79e-140
```




```
6LoWPAN BER at 20 m = 3.017266387017735e-19
```

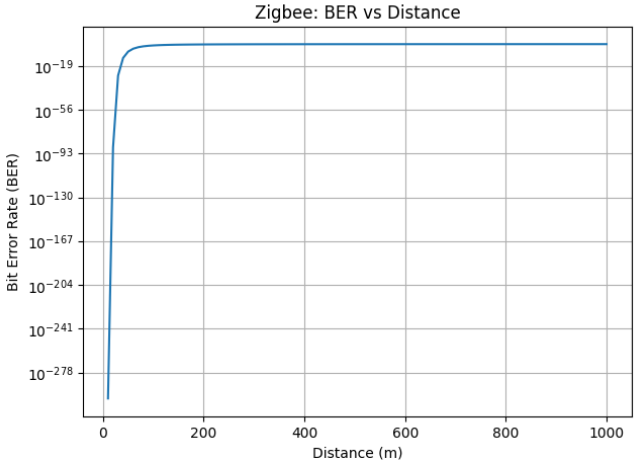


```
NB-IoT BER at 400 m = 0.1880965917311837
```

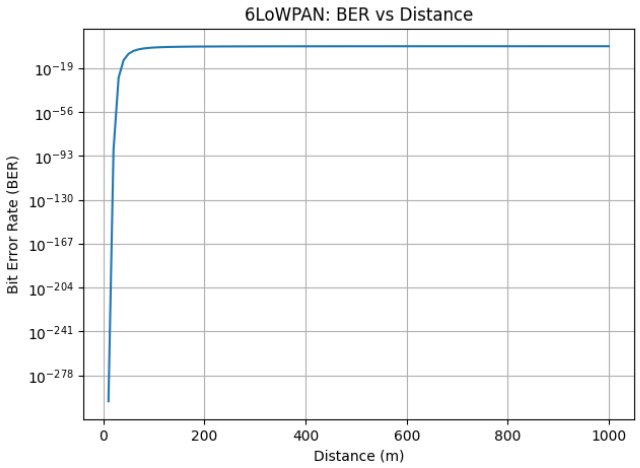


(iii) N = - 102

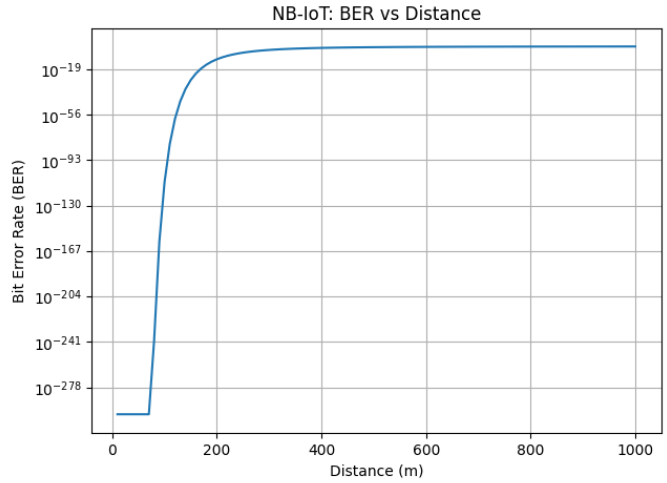
```
Zigbee BER at 10 m = 0.0
```




```
6LoWPAN BER at 20 m = 1.828330992115939e-88
```



```
NB-IoT BER at 400 m = 0.02378910568977974
```



Result :

The BER-versus-distance graphs are generated for all three protocols. With increasing distance, received signal strength and SNR decrease, generally causing BER to increase. Under the given simulation parameters, NB-IoT is expected to provide better long-range communication performance than Zigbee and 6LoWPAN.